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Cold chain transportation decision in the vaccine supply chain

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ABSTRACT

Vaccines are a special kind of drug, the quality of which is highly sensitive to temperature and directly related to public health. Recently, numerous vaccine-related adverse events have occurred in the world, especially in developing countries, due to vaccines being exposed to inappropriate temperatures during their transportation. This paper considers the vaccine supply chain including a distributor and a retailer (hospital or clinic). The distributor decides to use a cold chain or non-cold chain to transport the vaccines. The retailer performs an inspection when receiving the vaccines. First, a basic model is developed to study the conditions under which the distributor will transport the vaccines via a cold chain or non-cold chain. Then, two common inspection policies (a single-step one and a two-step one) are introduced into the basic model to explore the impact of the retailer's inspection at the end of transportation on the distributor's original decision. We show that the retailer's single-step inspection influences the distributor to choose the cold chain option. Interestingly, we prove that the two-step inspection policy is less effective than the single-step one in this effect. We suggest that the retailer's role in improving the distributor's non-cold chain transportation behavior should be fully used.

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1. Introduction

Vaccines are one of the most cost-effective methods to prevent infectious diseases. The World Health Organization (WHO, 2018) reported that vaccinations could prevent two to three million deaths per year. According to estimates, the mortality reductions in nine diseases due to vaccinations averaged 97.8% (Ventola, 2016). However, in recent years, quite a few vaccine-related adverse events put many recipients at risk, especially in developing countries. For example, in Indonesia, counterfeit polio vaccine production resulted in polio outbreaks in 2005 and 2016. In the Philippines, more than 730,000 children over the age of 9 were inoculated with dengue vaccines that were produced by Sanofi Pasteur in 2017, but the vaccines may be harmful to people who had not been previously infected. In China, Changsheng Biotechnology Company violated childhood vaccine production standards and counterfeited production records in 2018, which raised widespread concern in China. In Australia, vaccines were found to be incor-

rectly stored or expired at a practice in Sydney in 2019, which caused that 3000 patients who received vaccinations here since 2010 had to be vaccinated again.

Unlike other common drugs, vaccines are unique drugs, which are very sensitive to temperature. When vaccines are exposed to temperatures outside the appropriate range, their potency diminishes (WHO, 2015). That is, the vaccines become useless. Therefore, in order to maintain their quality, vaccines should be continuously kept within their determined temperature range from production to use. The lack of proper storage and transportation temperatures for vaccines is one of the common factors limiting full and equitable immunization in many countries (Brison & LeTallec, 2017). In this paper, we define a non-cold chain as that a company does not use a cold chain or that it uses a cold chain but fails to meet standards. Some stingy companies in the vaccine industry will use non-cold chains to store or transport vaccines. For instance, in 2016, 25 kinds of vaccines were found to be illegally distributed to medical facilities in at least 24 provinces in China without approved temperature conditions since 2011, causing economic losses of more than 88 million dollars (Cao, Zheng, Cao, Cui & Xiao, 2018; Qiu, Hu, Zhou & Liu, 2016).

A person is naturally protected with their immune system against diseases. Vaccines are an extra layer of protection. Some

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stingy companies will use non-cold chains assuming that the individual will not be infected and therefore will never discover useless vaccines. For example, less than 5% of healthy adults who are infected with hepatitis B will develop chronic infections (WHO, 2019). In other words, for healthy adults who are infected with hepatitis B, there is a small probability of developing chronic infections, even if the recipients receive useless vaccines. Most individuals do not have the means, knowledge, or capability to differentiate between effective and ineffective vaccines, which can only be done using special equipment and professional technicians. Additionally, many developing countries do not have sufficient cold chain capacity (Ashok, Brison & LeTallec, 2017) and effective vaccine regulation policies and penalties. Moreover, unreliable electricity power systems and poor road conditions in many developing countries often result in cold chain breakdowns (Duijzer, Jaarsveld & Dekker, 2018; Lauton, Rothkopf & Pibernik, 2019). The WHO and UNICEF (the United Nations International Children's Emergency Fund) assessments in 65 low and lower-middle income developing countries revealed that few countries met minimum standards for effective vaccine storage, distribution, handling, and stock management (Lydon, Raubenheimer, Arnot-Krüger & Zaffran, 2015). Hence, it is reasonable to deduce that the China's vaccine distribution scandal in 2016 is not an isolated case in developing countries. For instance, transportation of some kinds of vaccines requires refrigeration and freezing (Goldberg & Karhi, 2019). However, it was reported that 35.3% of shipments and 21.9% of refrigerators were found to be at temperatures below the WHO recommended freezing temperature range for vaccines (Matthias, Robertson, Garrison, Newland & Nelson, 2007). Murhekar et al. (2013) found that up to two-thirds of vaccines were damaged by freeze exposure in transit between state stores and administration sites across ten states in India and that exposure to subzero temperatures was frequent during vaccine storage at peripheral facilities and vaccine transportation.

Unsound vaccines that are caused by non-cold chain storage and transportation put a large population at long-term risk of potential outbreaks of some diseases. Thus, non-cold chain storage and transportation has drawn public attention and has taken a center stage of concern within the vaccine management research community. The increased awareness of the risks arising from unsound vaccines has prompted relevant studies. Duijzer et al. (2018) discuss 65 publications that deal with topics that are related to vaccination in top Operations Research/Operations Management journals. They classify those publications into the four groups of product, production, allocation, and distribution, and then they identify the promising research directions in this relatively new field. In our paper, we focus on the distribution part of the vaccine supply chain, since it is the longest portion of the transportation process and has many handling steps that are subject to inappropriate temperatures.

In this paper, we discuss the distributor's transportation decision to adopt a cold chain for vaccines or not. Then, we explore how the retailer's inspection at the end of transportation affects the distributor's decision at the beginning. We make the following contributions. First, we develop a basic model to discuss the distributor's cold chain transportation decision and the conditions under what the distributor will use a cold chain or non-cold chain for vaccines. Second, we show that the retailer's inspection at the end of transportation affects the distributor's decision at the beginning and influences the distributor towards the cold chain option (and away from the non-cold chain option). Third, the analysis of two commonly used inspection policies of the retailer shows that the two-step inspection policy that seems stricter than the single-step one is less effective in this influence. Overall, we illustrate the retailer's role in influencing the distributor's cold chain or non-cold chain decision and suggest that the retailer's role in improving the

distributor's non-cold chain transportation behavior should be fully used.

The remainder of this paper is organized as follows. In Section 2, we review the relevant literature. In Section 3, we establish the basic model of the distributor's cold chain transportation decision and discuss the threshold values of the key driver, the costs of a cold chain. In Section 4, we analyze how the retailer's inspection affects the distributor's decision-making. In Section 5, we conclude our work.

2. Literature review

Due to the significant role of vaccines in preventing the outbreaks of infectious diseases, researchers have been interested in and studied the vaccine supply chain from various perspectives. Lemmens, Decouttere, Vandaele and Bernuzzi (2016) review the relevant literature to determine whether the decisions at the strategic, tactical, and operational levels are able to address the vaccine supply chain's key issues, such as limited shelf life, cold chain distribution, and accessing remote areas. Dai, Cho and Zhang (2016) study a supply chain contracting problem considering the uncertainties surrounding the design, delivery, and demand of the influenza vaccine. They construct a buyback-and-late-rebate (BLR) contract in order to coordinate the supply chain and provide full flexibility for dividing the profits between the members of the supply chain. Chick, Hasija and Nasiry (2017) explore the government procurement of the influenza vaccine whose supplier is for-profit and has an uncertain production yield, private information, and potentially unverifiable production efforts. They provide the optimal menu within practically implementable contracts. Lee and Haidari (2017) indicate that the failure to understand and properly address issues in the vaccine supply chain will greatly reduce vaccines' effects. They discuss how the different roles in vaccine decision-making are affected by considering the vaccine supply chain's effects. Buyuktahtakin, des-Bordes and Kibiş (2018) introduce a new epidemics-logistics mixed-integer programming model to control an infectious disease outbreak. Duijzer, van Jaarsveld and Dekker (2018) model disease progression using the seminal SIR (Susceptible-Infected-Recovered) model and discuss the benefits of combining early aspecific vaccination with later specific vaccination. Shamsi, Ali Torabi and Shakouri (2018) use the SIR model to develop a contract for provisioning vaccines from two suppliers in order to ensure the timely and adequate supply of vaccines in disastrous situations. Rahimian, Bayraksan and Homem-de-Mello (2019) use distributionally robust optimization to control risk and demand ambiguity in newsvendor models that are fundamental to many operations models, such as vaccine production. Wu, Wang and Shang (2019) study the multi-sourcing and vertical information sharing problem in the supply chain where firms often employ multi-sourcing facing supply uncertainty, such as vaccine supply. These studies illustrate the importance of the vaccine supply chain to vaccines' effects.

Among the literature on the vaccine supply chain, vaccine quality issues are of particular relevance to our work. Crawford et al. (2014) compare the passive and active surveillance of adverse events following immunization (AEFI) and discuss the role of the active surveillance in vaccine safety programs. Liu et al. (2015) review the development, status, and key aspects of the Chinese AEFI surveillance system and describe the challenges and plans for vaccine safety assessments in China. Shimabukuro, Nguyen, Martin and DeStefano (2015) describe the fundamental vaccine safety concepts. They refer to a vaccine adverse event as an adverse event following immunization, i.e., an adverse health event or problem that occurs following or during the administration of a vaccine. Lopalco (2016) indicates that during the last decades, effective communication has become increasingly more impor-

tant due to the progressive lack of public confidence towards vaccination. Evidence-based communication that is supported by reliable information on vaccine effectiveness and safety may be central for improving vaccine confidence and assuring mutual protection. Clements, Lawrence and Macartney (2017) describe the efforts that have been taken to ensure that a vaccine is manufactured, tested, and administered within a safe environment and identify how vaccine safety is measured and monitored after a vaccine is licensed for use in the population. The research work on vaccine quality issues mainly focuses on the in-process surveillance and subsequent response of AEFI, while it lacks in-advance causal analysis and prevention. In addition, the focus of much of the existing work is on vaccine quality issues with respect to the vaccines themselves, while less attention is paid to the vaccine quality issues that are caused by external factors, such as non-cold chain transportation.

Although the existing literature on vaccine quality issues has yet to be supplemented, extensive work has been carried out on product quality issues in other settings. Villas-Boas (1998) models a product line design problem for the distribution channel where different products are identified by different quality levels and the customer market is composed of different segments that value quality differently. Wertheimer, Chaney and Santella (2003) examine the problem of drug counterfeiting and its effects around the world in order to consider the likely directions that this problem will take. Starbird and Amanor-Boadu (2008) believe that information asymmetry is one of the main causes for agricultural products' quality and safety problems. Xu (2009) discusses and compares the joint wholesale pricing and product quality decision problems of the manufacturer in two distribution channels considering different features of its marginal revenue function. Shi, Liu and Petruzzi (2013) study the optimal quality decision of the manufacturer with different distribution channel structures whose effect on product quality depends on the type of consumer heterogeneity and consumer distribution in the market. Degardin, Roggo and Margot (2014) show that medical counterfeiting is a serious worldwide issue involving manufacturing and distribution issues. The huge profits that are made by counterfeiters and the complexity of drug markets are the two main reasons for the expansion of this phenomenon. For instance, in 2007, Changzhou SPL, one of Baxter's contract manufacturers, used hypersulfated chondroitin sulfate to produce an adulterated blood-thinning drug, Heparin. In 2015, the New York Attorney General's Office exposed four retailers, Wal-Mart, GNC, Target, and Walgreens, for selling counterfeit dietary supplements. Eser, Kurtulusoglu, Bicaksiz and Sumer (2015) summarize the demand and supply sides of counterfeits and analyze the counterfeit supply chain in Turkey based on semi-structured interviews. Liu, Shi and Petruzzi (2018) analyze how market size uncertainty affects the effects of centralized and decentralized channel distribution on the manufacturer's optimal quality provision for the market where consumers are heterogeneous in valuing product quality, and then they prove that the market size uncertainty decreases the quality differential. Zhang, Cao and He (2019) analyze the interrelationship between an e-retailer platform's contract choice and a manufacturer's product quality decision. They find that product quality, whether exogenously or endogenously given, affects a platform's contract choice.

Some research discusses how to deal with product quality issues. Mackey and Liang (2011) propose a global policy framework utilizing public-private partnership (PPP) models with centralized surveillance for cooperation and coordination in order to combat the counterfeit drug industry. Babich and Tang (2012) compare three mechanisms for managing product adulteration problems: deferred payments, inspection, and combined mechanisms. Tang and Babich (2014) discuss how to use social and economic motives to reduce Chinese product adulteration. They first identify

four underlying motives: severe price pressure, short-term opportunism, asymmetric information, and rampant government corruption and ineffectual legal system. Zhang and Xue (2016) conduct an aggregated analysis on food fraud and economically motivated adulteration in China based on 1553 media reports on food safety scandals and concerns.

Our work is also related to the literature on the implication of the cold chain. Matthias et al. (2007) point out that the specificity of vaccine management puts additional pressure on the already fragile cold chain, the distribution network, and the procedure that is used to maintain vaccine quality from the manufacturer to recipients. Due to the temperature sensitivity of biopharmaceuticals, the cold chain has become an increasingly significant part of the overall pharmaceutical supply chain (Bishara, 2006). Cai, Chen, Xiao and Xu (2010) consider a long-distance transportation supply chain in which the distributor procures a kind of fresh product from the producer and then has to make an appropriate effort to preserve the freshness of the products. A model is developed considering factors including the level of the freshness keeping effort and the selling price that is affected by the freshness. Lan, Zhao, Su and Liu (2014) analyze the food cold chain equilibrium based on the collaborative replenishment policy. The supplier and the retailer participate in the non-cooperative game to achieve the equilibrium in terms of quantity and price while considering the relationship between food quality and its price. Yu and Xiao (2017) develop two Stackelberg models to investigate the pricing and service level decisions of a fresh agri-product supply chain consisting of a supplier, a retailer, and a third-party logistics provider while considering the exogenous cold chain service price. Hibbs et al. (2018) search the VAERS database from 2008–2012 for reports describing vaccines being kept outside the recommended temperature range and analyze those reports in order to determine whether cold chain management breakdowns will make vaccines unsound. They suggest that the lack of vigilance, inadequate training, and equipment failure are the reasons that are cited for cold chain management breakdowns. The literature in this area mainly discusses two topics of the cold chain. One is the significance of the cold chain to temperature-sensitive products, and the other is the decision models regarding the cold chain as an exogenous variable that impacts product quality. Our work differs from the previous literature by focusing on the cold chain transportation decision in the vaccine supply chain and taking its impact on vaccine quality into account.

Overall, scholars study the topics related to the vaccine supply chain, vaccine and product quality issues, and the implication of the cold chain. Their work provides ideas and methods that are helpful to our work. However, the existing research on vaccine quality issues and the vaccine cold chain is limited. To the best of our knowledge, our work is the first one to discuss the distributor's cold chain transportation decision. Our work studies under what conditions the distributor will use a cold chain or non-cold chain to transport vaccines considering the impact of a cold chain on vaccine quality. Further, we introduce the role of the retailer into the basic model. We explore how the retailer's inspection at the end of transportation affects the distributor's original decision.

3. Distributor's cold chain transportation decision

3.1. Problem description

The vaccine supply chain in this paper includes a distributor and a retailer (hospital or clinic). We model the situation where the for-profit distributor is the intermediary that purchases temperature-sensitive vaccines from the manufacturer and then resells and transports them to the retailer in developing countries, such as India (Hagan et al., 2018), Nigeria (Aina et al., 2017). This section focuses on the distributor's decision of its transportation

Table 1
Probabilities of the transportation situations.

Distributor's decision	Transportation situation	Probability of the situation
$t = C$	Both sets of equipment work well	$(1 - \rho_1)(1 - \rho_2)$
	Both sets of equipment malfunction	$\rho_1 \rho_2$
	Only the controlling equipment malfunctions	$\rho_1(1 - \rho_2)$
	Only the recording equipment malfunctions	$(1 - \rho_1)\rho_2$
$t = NC$	/	/

Table 2
Probabilities of vaccines belonging to a certain type in the different situations.

Distributor's decision	Transportation situation	Probability of vaccines belonging to a certain type	
		Valid	Invalid
$t = C$	Both sets of equipment work well	α	$1 - \alpha$
	Both sets of equipment malfunction	0	1
	Only the controlling equipment malfunctions	0	1
	Only the recording equipment malfunctions	α	$1 - \alpha$
$t = NC$	/	0	1

mode, using a cold chain or non-cold chain. As mentioned in the Introduction section, the definition of a non-cold chain is that the distributor does not use a cold chain or that it uses a cold chain but fails to meet standards.

Vaccines have a potency attribute that is represented by θ_i , where $i \in \{V, I\}$ represents the type of vaccines: *valid* (V) or *invalid* (I). Specifically, θ_i is the probability of an adverse event related to vaccines whose type is i . Apparently, valid vaccines have a lower adverse event probability, i.e., $0 < \theta_V < \theta_I < 1$. Assume that the manufacturer produces vaccines that are up to standards. However, due to the inherent variability of biological processes, vaccines still have invalidation risks with the probability $1 - \alpha$. That is, in the beginning, the type of vaccines is valid with the prior probability α , and it is invalid with the complementary probability $1 - \alpha$. The actual initial type of the vaccines is unknown, but both θ_i and α are common knowledge. To simplify, assume that the type of vaccines is only affected by the distributor after being produced, or more specifically, its transportation mode. This precondition ensures that our work concentrates on the distributor's decision with respect to its transportation mode, using a cold chain or non-cold chain.

We next describe the decision process of the distributor. The distributor has two options of its transportation mode that is denoted as $t \in \{C, NC\}$, where C represents *cold chain*, and NC represents *non-cold chain*. The costs of a cold chain k are exogenous, which is similar to that in Blackburn and Scudder (2009), Saif and Elhedhli (2016), and Yu and Xiao (2017). Core equipment in a cold chain is temperature controlling equipment and temperature recording equipment (Bogataj, Bogataj & Vodopivec, 2005; WHO, 2015), while malfunctions of which are common and result in temperature damage to vaccines (Ashok et al., 2017; Hibbs et al., 2018). For example, thousands of doses of the H1N1 vaccine were lost due to a refrigerator malfunction at a Pennsylvania public school in 2009 (Rogers, Dennison, Adepoju, Dowd & Uedoi, 2010).

The malfunction probabilities of both sets of equipment are ρ_1 and ρ_2 , respectively. Therefore, there are five possible transportation situations for vaccines. The first is that both sets of equipment work well, the second is that both sets of equipment malfunction, the third is that only the controlling equipment malfunctions, the fourth is that only the recording equipment malfunctions, and the fifth is that the distributor uses a non-cold chain for vaccines. The probabilities of these situations occurring are shown in Table 1.

As shown in Table 2, these five transportation situations correspond to different probabilities of vaccines belonging to a certain type. We define the vaccines we discuss in this paper as temperature-sensitive vaccines, such as hepatitis B vaccines that

are very freeze sensitive, oral polio vaccines that are very heat sensitive, and DTaP vaccines that are sensitive both to freeze and to heat. Hence, if the controlling equipment keeps working during transportation, the type of the vaccines will remain unchanged; otherwise, the vaccines will lose their potency. That is, the vaccines will be definitely invalid, which is the same as that when the distributor uses a non-cold chain.

3.2. Notations

The notations used in developing the basic model are given as follows:

i	type of vaccines, where $i = V$ for valid vaccines and $i = I$ for invalid vaccines
θ_i	probability of an adverse event related to vaccines whose type is i
α	prior probability that vaccines are valid
k	costs of a cold chain
ρ_1	malfunction probability of the temperature controlling equipment
ρ_2	malfunction probability of the temperature recording equipment
r	transportation result, where $r = D$ for data and $r = ND$ for no data
λ	retailer's trust level in the distributor to use a cold chain, $\lambda \in [0, 1]$
μ	retailer's belief about valid vaccines, related to r and λ , $\mu \in [0, 1]$
ν	retailer's income related to vaccines
L	loss of a vaccine-related adverse event, which is borne by the distributor and the retailer
L_D	distributor's loss in a vaccine-related adverse event
L_R	retailer's loss in a vaccine-related adverse event
p	vaccine price that the distributor and the retailer trade at
η	ratio of the distributor's loss to the full loss of a vaccine-related adverse event, $\eta = L_D/L$
Δ	difference between the adverse event probabilities of the two types of vaccines, $\Delta = \theta_I - \theta_V$

Decision variable

t	Distributor's transportation mode, where $t = C$ represents cold chain and $t = NC$ represents non-cold chain
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3.3. Basic model

There are five possible transportation situations for vaccines. Only in the first situation can the distributor obtain the

necessary temperature data, which means that the data are complete and within the regulated range. In the other four situations, the data are incomplete, some of them are outside the regulated range, or, even worse, there is no data record when the distributor forgoes the use of a cold chain. The transportation result that is either *data* or *no data* is denoted as $r \in \{D, ND\}$. Let λ represent the trust level of the retailer in the distributor to use a cold chain, based on which the retailer forms the belief about valid vaccines $\mu(r, \lambda)$ using Bayes' rule. Given λ , when $r = D$, $\mu(D, \lambda) = \alpha$, and when $r = ND$, $\mu(ND, \lambda) = \frac{(1-\rho_1)\rho_2\lambda}{[1-(1-\rho_1)(1-\rho_2)]\lambda + 1 - (1-\lambda)}\alpha$. In terms of $\mu(ND, \lambda)$, the denominator is the probability of the transportation result being no data, and the numerator is the retailer's expected probability of the situation where only the recording equipment malfunctions occurring. Among the no data situations, only in the recording equipment malfunction situation can the vaccines keep their effectiveness. Hence, the fraction and the prior probability of vaccines being valid compose the retailer's belief about valid vaccines when the transportation result is no data.

The expected adverse event probability of the retailer is $\mu\theta_V + (1-\mu)\theta_I$. The first part represents the expected adverse event probability when the retailer considers the vaccines to be valid with its belief μ , and the second part represents the expected adverse event probability with the retailer's belief that the vaccines are unsound $1-\mu$. Let L represent the full loss of a vaccine-related adverse event. L is borne by the distributor (L_D) and the retailer (L_R), namely, $L = L_D + L_R$. Hence, the expected loss of the retailer is $[\mu\theta_V + (1-\mu)\theta_I]L_R$. Let v represent the income of the retailer that is related to vaccines and p represent the vaccine price that the distributor and the retailer trade at. We model the retailer's payoff structure as follows:

$$\pi^R(\mu, p) = v - p - [\mu\theta_V + (1-\mu)\theta_I]L_R \quad (1)$$

In many areas of developing countries, the cold chain transportation industry has not been well established yet, and there are few qualified distributors. Hence, we regard a local vaccine distribution market as a monopoly market. The vaccine price extracts all of the retailer's surplus profits, which means that it is the highest trading price that the retailer will accept. The retailer has a minimum profit target, which is a fixed amount. To keep results and discussion concise, we assume that the retailer's surplus profits are zero. We can infer the following:

$$p = v - [\mu\theta_V + (1-\mu)\theta_I]L_R \quad (2)$$

The vaccine price p is related to the retailer's belief about valid vaccines μ , which is formed based on the transportation result r and the retailer's trust level λ . Therefore, the distributor and the retailer will trade at two different prices concerning two possible transportation results. If the distributor provides the data to the retailer on delivery, they trade at price p_D , and, if not, they trade at price p_{ND} , which is less than p_D . It should be noticed that p is also taken as the maximum price that the retailer will accept under the specified transportation results.

For the distributor, the expected adverse event probability is $(1-\rho_1)\alpha\theta_V + [1-(1-\rho_1)\alpha]\theta_I$ when it decides to use a cold chain ($t = C$), or it is θ_I in the other case ($t = NC$). Based on the discussion above, we infer Proposition 1 describing the distributor's cold chain transportation decision (Proof: see the Appendix).

Proposition 1. *In the basic model, the distributor chooses its transportation mode according to the costs of a cold chain as follows.*

- (1) If $k \leq k(\lambda)$, the distributor uses a cold chain to transport vaccines; and
- (2) If $k > k(\lambda)$, the distributor uses a non-cold chain to transport vaccines,

$$\text{where } k(\lambda) = (1-\rho_1)\alpha(\theta_I - \theta_V) \left\{ \frac{(1-\rho_2)[1-(1-\rho_1)\lambda]}{1-(1-\rho_1)(1-\rho_2)\lambda} L_R + L_D \right\}.$$



Fig. 1. Distributor's decision in the basic model.

Since the first derivative of $k(\lambda)$ with respect to λ is negative, we infer that the more the retailer trusts the distributor, the smaller the distributor's motive to use a cold chain. The derivation process of Proposition 1 specifically shows that the distributor's motive to use a cold chain strictly decreases with the decreasing price difference $p_D - p_{ND}$. Further, the price difference strictly decreases with the increasing trust level of the retailer in the distributor to use a cold chain λ .

Case study. In March 2016, a vaccine scandal that was uncovered in Shandong province, China infuriated the public and damaged confidence in immunization. The prime culprits were arrested reselling and transporting vaccines to medical facilities in at least 24 provinces without approved temperature conditions since 2011, involving 25 kinds of problematic vaccines, including polio, mumps, rabies, hepatitis B, encephalitis, and meningococcal vaccines (Qiu et al., 2016). This scandal motivates our paper to explore why the distributor uses a non-cold chain for vaccines opportunistically. Here, we provide a hypothetical case study using parameter values that are estimated from some literature.

For the hepatitis B vaccine, a safe and effective vaccine will offer a 98–100% protection against hepatitis B, and less than 5% of healthy adults who are infected with hepatitis B will develop chronic infections (WHO, 2019). For the influenza vaccine, Weycker et al. (2005) assume the vaccine efficacy as 70% among children and younger adults, and 50% among older adults. Sugaya et al. (2018) show that the adjusted vaccine effectiveness of the quadrivalent inactivated influenza vaccine using during 2015/16 season in Japan was 49% against any type of influenza, 57% against influenza A, and 34% against influenza B, given the 95% confidence interval. Similar studies have been taken for other countries, such as China (Wu et al., 2018), Singapore (Ng et al., 2019), the United States (Shang et al., 2018). Here, we take the hepatitis B vaccine as an example and set $\theta_V = 2\%$ and $\theta_I = 5\%$.

China Food and Drug Administration reports that the production qualification rate of vaccines during 2008 to 2017 was up to 99.6%. Preiss, Garçon, Cunningham, Strugnell and Friedland (2016) indicate that nearly 70% of vaccine production time is dedicated to quality control and confirmation measures. Hence, we assume $\alpha \in [0.7, 1]$ and choose $\alpha = 0.7$ in this case. Further, we set $\rho_1 = 33\%$ from the break frequency in cold chain or refrigeration in Setia et al. (2002) and $\rho_2 = 41\%$ from the approximate frequency of the temperature not being recorded twice daily in Ateudjieu, Kenfack, Nkontchou and Demanou (2013).

Weycker et al. (2005) and Chick, Mamani and Simchi-Levi (2008) estimate the average direct costs of each infected individual over different groups with 95. Chick et al. (2008) take the average cost per infected individual between [70, 120] in their experiment. We set $L = 95$. According to the retail price of the adult hepatitis B vaccine (Centers for Disease Control & Prevention, 2019), we set $v = 50$.

Fig. 2 shows the threshold value of the cold chain costs $k(\lambda)$ changing with the retailer's trust level of the distributor to use a cold chain for vaccines λ . For instance, when $\lambda = 0.5$ and $\eta = 0.6$, $k(0.5) = 1.06$. If the distributor can make the costs less than 1.06, it will use a cold chain; otherwise, it will use a non-cold chain, even though the costs are only around 1.12% of the loss of a vaccine-related adverse event. Additionally, $k(\lambda)$ increases with η .

Further, we discuss two extreme values of $k(\lambda)$. First, we get $k' = k(1) = (1-\rho_1)\alpha(\theta_I - \theta_V) \left[\frac{(1-\rho_2)\rho_1}{\rho_1 + \rho_2 - \rho_1\rho_2} L_R + L_D \right]$, which is the threshold value of the cold chain costs when the retailer entirely

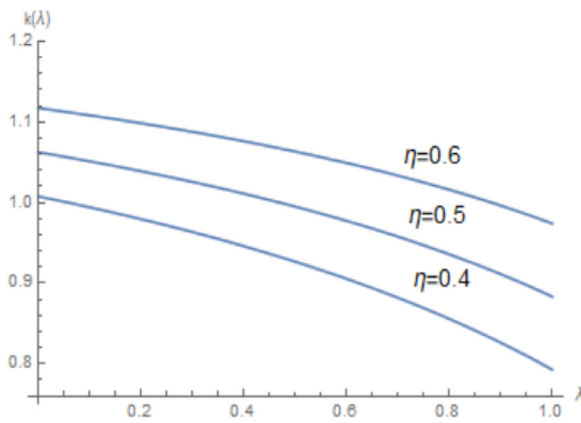


Fig. 2. Distributor's decision in case study.

trusts the distributor ($\lambda = 1$). Even in this case, the distributor will use a cold chain if $k \leq k'$; otherwise, it will use a non-cold chain to save the costs k and meanwhile trade with the retailer at the same price as that in the malfunctioning transportation situations, which is higher than the actual non-cold chain vaccine price. Second, we get $k'' = k(0) = (1 - \rho_1)\alpha(\theta_l - \theta_v)[(1 - \rho_2)L_R + L_D]$, which is the threshold value when the retailer does not trust the distributor at all ($\lambda = 0$). As mentioned, $k(\lambda)$ increases with the decrease of λ . Hence, no matter what value λ takes, as long as $k \leq k'$, the distributor will choose the cold chain option, and, as long as $k > k''$, it will choose the non-cold chain option.

Proposition 2. No matter what value λ takes, we have the following results.

- (1) If $k \leq k'$, the distributor uses a cold chain to transport vaccines; and
- (2) If $k > k''$, the distributor uses a non-cold chain to transport vaccines.

The summary of the distributor's decision in the basic model is shown in Fig. 4, where the shadow area represents cold chain and the rest represents non-cold chain. When $k < k'$, the costs of a cold chain are small enough to ensure that the distributor will use it; when $k > k''$, the costs are so high that the distributor will not benefit from using a cold chain; and when $k' < k < k''$, the distributor's decision is related to the trust level of the retailer in the distributor to use a cold chain.

Note that sometimes when a vaccine-related adverse event occurs, the distributor can be exempt from liabilities if the transportation result is data. In this situation, $k(\lambda)$, k' and k'' will move to the left side with the exemptible part of liabilities $(1 - \rho_1)(1 - \rho_2)[\alpha\theta_v + (1 - \alpha)\theta_l]L_D$. Since the situation does not have a significant effect on our analysis, the work is conducted based on the upper basic model hereafter.

Next, we focus on the ratio of the distributor's loss to the full loss of a vaccine-related adverse event $\eta = L_D/L$ and the difference between the adverse event probabilities (the potencies) of the two types of vaccines $\Delta = \theta_l - \theta_v$. We show that the ratio and the difference both affect the distributor's decision (Proof: see the Appendix).

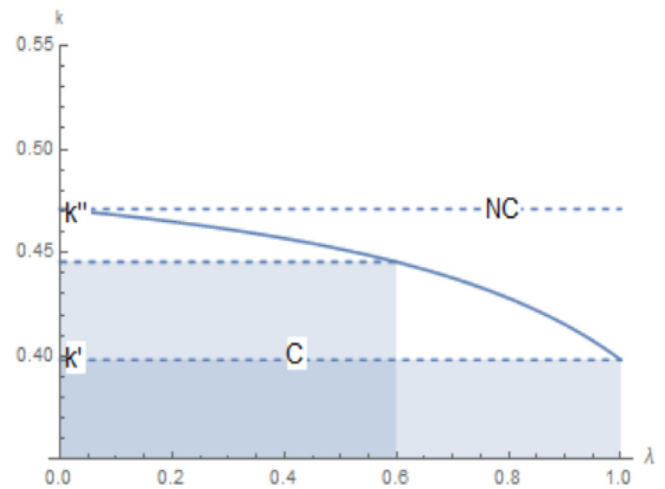
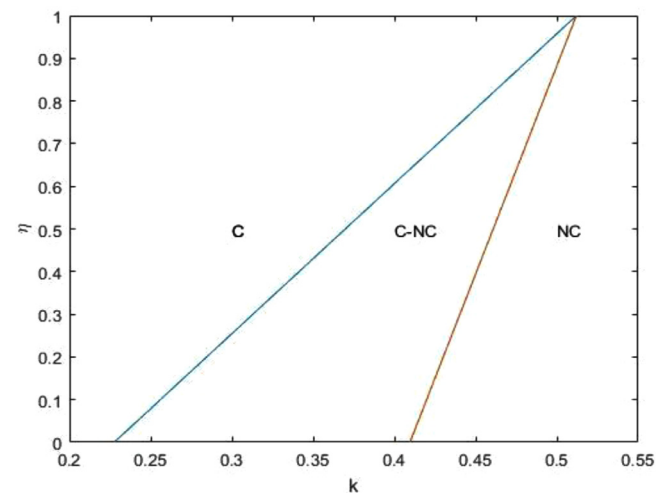


Fig. 4. Summary of the distributor's decision in the basic model.

Notes: Example for $\alpha = 0.8$, $\rho_1 = \rho_2 = 0.2$, $L_D = 1.2$, $L_R = 0.8$, $\theta_l = 0.5$, $\theta_v = 0.1$.

Fig. 5. Distributor's decision in the $k - \eta$ space in the basic model.

Notes: Example for $\alpha = 0.8$, $\rho_1 = \rho_2 = 0.2$, $L = 2$, $\theta_l = 0.5$, $\theta_v = 0.1$.

Proposition 3. The threshold values of the distributor's decision are affected by the ratio and the difference.

- (1) $k(\lambda)$ increases strictly in η and Δ ; and
- (2) k' and k'' increases strictly in η and Δ .

Part (1) of Proposition 2 shows that the higher the ratio and the difference, the wider the tolerance range of the cold chain costs for the distributor. Intuitively, the more loss the distributor has to bear in a vaccine-related adverse event, the greater motive it has to use a cold chain. For vaccines that have higher potency, the distributor is more willing to use a cold chain. Part (2) of Proposition 2 says that increases in η and Δ also increase the extreme threshold values, which is socially beneficial. These results are represented as rightward shifts of $k(\lambda)$, k' and k'' in Fig. 3 and are also presented in Figs. 5 and 6.

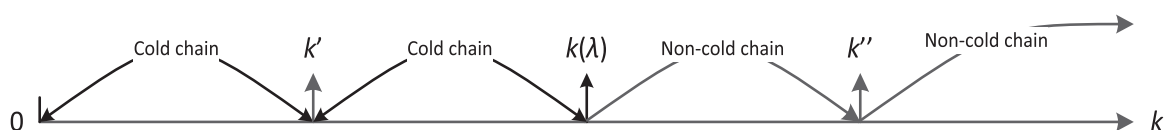


Fig. 3. Extreme threshold values in the basic model.

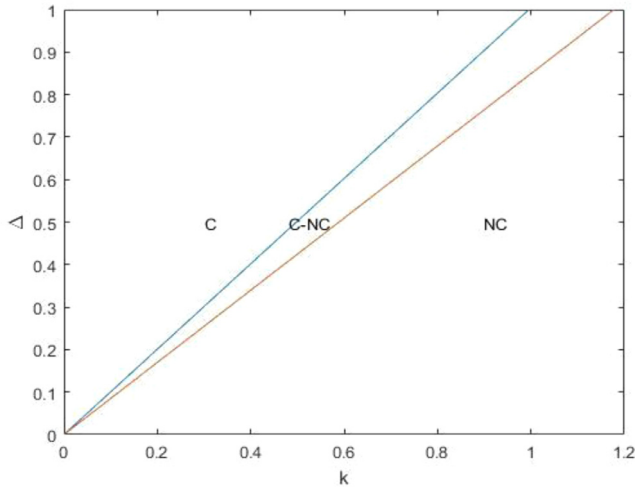


Fig. 6. Distributor's decision in the $k - \Delta$ space in the basic model.
Notes: Example for $\alpha = 0.8$, $\rho_1 = \rho_2 = 0.2$, $\eta = 0.6$, $L = 2$.

Fig. 5 says that as η increases, the proportion of the cold chain decision in the $k - \eta$ space is increasingly important, while the proportion of the non-cold chain decision decreases. This result appeals to common sense that with great responsibility, the distributor will curb its motive to choose the non-cold chain option. Therefore, a reason for the distributor to use a non-cold chain is that the penalties that the distributor will bear in a vaccine-related adverse event, such as governmental penalty, are not enough. Fig. 6 indicates the similar trend as Δ changes. For vaccines whose potency is significantly different between the two types, the exposure probability of the improper transportation is high. Hence, the distributor's motive to use a non-cold chain decreases. Based on the discussion above, Proposition 3 provides two managerial insights for policy makers. First, if feasible, suitably increasing the ratio η will prompt the distributor to choose the cold chain option and meanwhile reduce its motive to choose the non-cold chain option. Second, similar effects can be achieved by increasing the difference Δ . Policy makers should take measures that will improve vaccines' potency, such as investing in relevant research.

4. Impact of the retailer's inspection policy on the distributor's decision

Without an additional inspection, it is hard for the retailer to identify the distributor's decision regarding its transportation mode and the malfunctions of the cold chain. This inability motivates the distributor to use a non-cold chain opportunistically. In this section, we introduce the retailer's inspection policy into the basic model and study its impact on the distributor's decision-making.

The timing of their decisions is sequential. The retailer first determines its inspection policy and announces it to the distributor; and then the distributor makes its decision according to the costs of a cold chain as before. We use $i \in \{P, F\}$ to denote the inspection result, where P represents *pass* and F represents *fail*. Similarly, based on the inspection result i and the retailer's trust level in the distributor to use a cold chain λ , the retailer forms the belief about valid vaccines $\mu(i, \lambda)$ using Bayes' rule. Recall from Section 3 that the expected adverse event probability of the retailer is $\mu\theta_V + (1 - \mu)\theta_I$. Therefore, the retailer's payoff structure and the vaccine price in this section are as follows:

$$\pi^R(\mu, p) = v - p - [\mu\theta_V + (1 - \mu)\theta_I]L_R - \tau \quad (3)$$

$$p = v - [\mu\theta_V + (1 - \mu)\theta_I]L_R - \tau \quad (4)$$

Per the discussion above, the inspection result affects the retailer's belief about valid vaccines and further the vaccine price, which is the point that connects the retailer's inspection policy and the distributor's decision.

In what follows, we analyze two commonly used inspection policies of the retailer (a single-step one and a two-step one) in order to explore how the retailer's inspection policy influences the distributor's decision and which inspection policy is more effective in this influence. The single-step one is that the retailer examines whether the receiving temperature of the refrigerator car conforms to standards with the inspection costs τ . The two-step one is that on the basis of the single-step inspection policy, if the inspection result is pass, the retailer further checks whether the temperature data are complete and within the regulated range with the extra inspection costs $\Delta\tau$. Theoretically, the retailer must check two characteristics of the temperature data: the integrity and the range. However, if some temperature spots are outside the range, the distributor will remove them from the data file. Hence, we simplify the retailer's inspection steps to checking the integrity with an implicit precondition that the data are within the regulated range as long as they are provided. In this paper, we regard the inspection costs as exogenous, which is similar to that in Rui and Lai (2015) and Hassin and Roet-Green (2017). The inspection results in the different transportation situations under the two inspection policies are explained in detail later.

4.1. Single-step inspection policy

When the retailer chooses the single-step inspection policy, it will find that the receiving temperature is either within or out of the regulated range. Based on the discussion in Section 3, there are five possible transportation situations for vaccines. If the distributor uses a cold chain for vaccines, we can deduce that as long as the temperature controlling equipment operates well, the receiving temperature will pass the inspection. Once a malfunction occurs in the controlling equipment, the distributor will fail the inspection according to the probability β and pass with the complementary value $1 - \beta$, since the malfunction may occur and be addressed before arriving at the retailer. If the distributor uses a non-cold chain for vaccines, it will definitely fail the inspection. Based on the discussion above and the possible inspection results (see in Table 3), we introduce the following proposition (Proof: see the Appendix).

Proposition 4. When the distributor observes that the retailer chooses the single-step inspection policy, the following will result:

- (1) If $k < k_1 = (1 - \rho_1)\alpha(\theta_I - \theta_V)L$, the distributor uses a cold chain for vaccines; and
- (2) If $k > k_1 = (1 - \rho_1)\alpha(\theta_I - \theta_V)L$, the distributor uses a non-cold chain for vaccines.

It is obvious that k_1 is greater than $k(\lambda)$. Namely, the threshold value of the distributor's decision increases due to the retailer's single-step inspection policy. The difference between k_1 and $k(\lambda)$ is $\frac{(1-\rho_1)\rho_2}{[1-(1-\rho_1)(1-\rho_2)]\lambda+1*(1-\lambda)}\alpha(\theta_I - \theta_V)L_R$. Referring to the fraction, we regard the numerator as the probability of the situation where only the recording equipment malfunctions; and we regard the denominator as the probability of the situations where the transportation result is no data, given the retailer's trust level in the distributor to use a cold chain λ . Hence, we suggest that the difference is derived from the retailer partly separating the recording equipment malfunction situation from the other no data situations. Comparing k_1 with k' , we find that k_1 is greater than k' . When the retailer totally trusts the distributor ($\lambda = 1$), it will consider the probability of the situations where the transportation result is no data as $\rho_1 + \rho_2 - \rho_1\rho_2$. Now, the difference becomes $\frac{(1-\rho_1)\rho_2}{1-(1-\rho_1)(1-\rho_2)}\alpha(\theta_I - \theta_V)L_R$, that is the difference between k_1 and

Table 3
Inspection results under the retailer's single-step inspection policy.

Distributor's decision	Transportation situation	Inspection result
$t = C$	Both sets of equipment work well	P
	Both sets of equipment malfunction	P/F
	Only the controlling equipment malfunctions	P/F
	Only the recording equipment malfunctions	P
$t = NC$	/	F

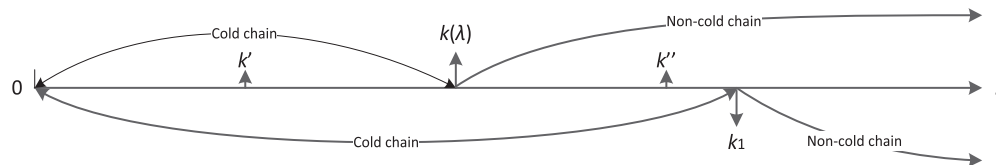


Fig. 7. Distributor's decision under the retailer's single-step inspection policy.

Table 4
Inspection results under the retailer's two-step inspection policy.

Distributor's decision	Transportation situation	Inspection result
$t = C$	Both sets of equipment work well	(P, P)
	Both sets of equipment malfunction	$(P, F)/F$
	Only the controlling equipment malfunctions	$(P, F)/F$
	Only the recording equipment malfunctions	(P, F)
$t = NC$	/	F

k' . Likewise, k_1 is also greater than k'' , which means that the possibility of the distributor definitely choosing the non-cold chain option decreases. The discussion above is intuitively presented in Fig. 7.

Interestingly, the threshold value of the distributor's decision is independent of the retailer's trust level in the distributor to use a cold chain. This happens because once the inspection fails, the retailer's belief about valid vaccines is zero, regardless of whether the distributor's decision is cold chain or non-cold chain. In other words, the inspection result offers additional information about the vaccines for the retailer. Note that the threshold value has nothing to do with the loss ratio η , while it shows a positive correlation with the difference between the adverse event probabilities of the two types of vaccines Δ . This result provides a new perspective for policy makers. In practice, it is usually hard to adjust the loss ratio due to the different interests and demands throughout the supply chain. Policy makers should bring the retailer into full play. The influence rate of Δ on the threshold value is $(1 - \rho_1)\alpha L$. Thus, the more effective vaccines are, the more effective the retailer's single-step inspection policy will be.

4.2. Two-step inspection policy

When the retailer chooses the two-step inspection policy, it will first inspect the receiving temperature; and if the inspection result of the first step is pass, the retailer will further check the integrity of the temperature data. There may be two inspection steps, and so the possible inspection results consist of (P, P) , (P, F) and F . Apparently, only when both sets of equipment function well can the result be (P, P) . If only the recording equipment breaks down, the result will be (P, F) . Once the controlling equipment malfunctions, the result will be (P, F) with the probability $1 - \beta$ and F with the probability β . When the distributor uses a non-cold chain for vaccines, the result will definitely be F . The possible inspection results are shown in [Table 4](#). The following proposition describes the distributor's decision under the retailer's two-step inspection policy (Proof: see the Appendix).

Proposition 5. *When the distributor observes that the retailer chooses the two-step inspection policy, we get the following:*

- (1) If $k < k_2 = (1 - \rho_1)\alpha(\theta_I - \theta_V)L - (1 - \rho_1\beta)\Delta\tau$, the distributor uses a cold chain to transport vaccines; and
- (2) If $k > k_2 = (1 - \rho_1)\alpha(\theta_I - \theta_V)L - (1 - \rho_1\beta)\Delta\tau$, the distributor uses a non-cold chain to transport vaccines.

Comparing [Proposition 5](#) to [Proposition 1](#), we get the following results.

When $\Delta\tau > \frac{(1-\rho_1)\rho_2}{(1-(1-\rho_1)(1-\rho_2)\lambda)(1-\rho_1\beta)}\alpha(\theta_I - \theta_V)L_R$, $k_2 < k(\lambda)$, which means the two-step inspection has a negative effect on the distributor's decision (shown in Fig. 8).

When $\Delta\tau < \frac{(1-\rho_1)\rho_2}{(1-(1-\rho_1)(1-\rho_2)\lambda)(1-\rho_1\beta)}\alpha(\theta_I - \theta_V)L_R$, $k_2 \geq k(\lambda)$, which means the two-step inspection is effective at influencing the distributor towards the cold chain option (shown in Fig. 9).

Comparing Proposition 5 to Proposition 4 leads to an interesting result. The two-step inspection policy causes the threshold value to shift left, i.e., $k_2 < k_1$. In other words, the two-step inspection policy is less effective than the single-step one at influencing the distributor to use a cold chain (shown in Fig. 10).

We suggest two explanations for this phenomenon. On the one hand, though the two-step inspection policy will recognize the situation where both sets of equipment work well, it then raises the vaccine price from p_P to $p_{(P,P)}$, which will motivate the distributor to choose the cold chain option. However, such an inspection will lead to a decrease of the vaccine price from p_P to $p_{(P,F)}$ in the other pass situations under the single-step inspection policy. As a result, the two opposite effects may cancel each other out. On the other hand, the inspection costs increase, which would force the retailer to shift the increasing costs to the vaccine price. Hence, it is understandable that smaller profits will enhance the distributor's motive to choose the non-cold chain option. Comparing these two inspection policies shows that the key point of the retailer's inspection policy is to improve its ability to identify the situation where only the recording equipment malfunctions from the other situations. However, further inspecting the temperature data cannot make the retailer achieve a better ability.

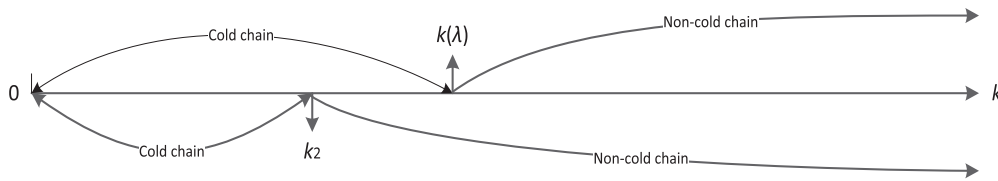
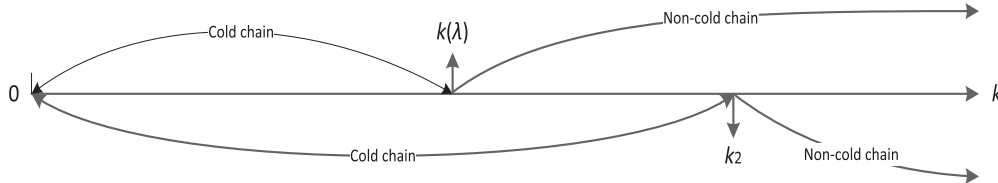
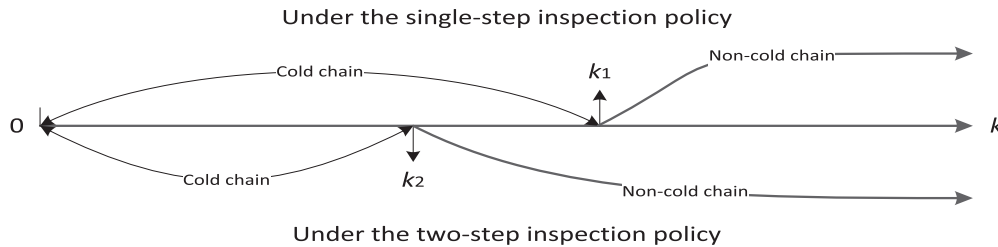
Fig. 8. Comparison of Proposition 5 to Proposition 1 with a big $\Delta\tau$.Fig. 9. Comparison of Proposition 5 to Proposition 1 with a small $\Delta\tau$.

Fig. 10. Comparison of the two inspection policies.

Regarding Proposition 4, a proper inspection of the retailer is effective and imperative for social welfare. However, a comparison of Proposition 5 and Proposition 4 indicates that the retailer's two-step inspection policy is less effective than the single-step one in improving the distributor's non-cold chain transportation behavior. The inadequate two-step inspection policy is wasteful, even though it seems stricter than the single-step one. Consequently, policy makers should guide the retailer that uses the two-step inspection policy to redesign an inspection policy and improve the ability to separate different transportation situations.

4.3. Discussion

In this section, we discuss two commonly used inspection policies (the single-step one and the two-step one). The single-step inspection policy has lower inspection costs than the two-step one. First, we prove that the retailer's single-step inspection policy influences the distributor towards the cold chain option and away from the non-cold chain option. Then, we compare these two inspection policies and find that even though the two-step inspection policy seems stricter than the single-step one, it is less effective in this influence (shown in Table 5).

In the basic model, the threshold value of the distributor's decision is $k(\lambda)$; and under the single-step inspection policy, the threshold value is k_1 ; and under the two-step inspection policy, the threshold value is k_2 (shown in Table 6).

Now we discuss the inspection policies in terms of the retailer, given two information situations of the cold chain costs: the costs are common knowledge of the distributor and the retailer, or the costs are private information of the distributor.

We first analyze the common knowledge situation. The retailer first declares which inspection policy it will perform. Since the retailer is aware of the distributor's cold chain costs, it will deduce the distributor's reaction to the inspection policy according to the propositions that we propose in the previous analyses, and then adjust its behavior, which further affects the distributor's decision. Under the single-step inspection policy, the

distributor should have used Proposition 4 to make its decision. That is, when $k < k(\lambda) < k_1$, it is reasonable for the retailer to suppose that the distributor will use a cold chain. Then, the retailer will not perform the inspection to save the inspection costs, which will not change the distributor's decision of using a cold chain since $k < k(\lambda)$ according to Proposition 1. Hence, cold chain and no inspection will be realized in this case. When $k(\lambda) < k \leq k_1$, the distributor should use a cold chain according to Proposition 4. Based on this deduction, the retailer will not perform the inspection, which then makes the distributor turn to the non-cold chain option according to Proposition 1. When $k > k_1$, the distributor will definitely choose the non-cold chain option, which means that it is not necessary for the retailer to perform the inspection. Similarly, we find that the retailer has no motive to perform the two-step inspection. Overall, the retailer will not perform an inspection, and the distributor will act according to Proposition 1, no matter which inspection policy the retailer declares to perform at the beginning. The retailer's announcement is incredible in the common knowledge situation. In this case, it is simply not enough to depend on the retailer to reduce the occurrence of vaccine-related adverse events resulting from non-cold chain transportation. External forces should play a role in regulating the retailer's inspection behavior.

For the private information situation, due to the lack of the information about the cold chain costs, the retailer will perform the single-step inspection or the two-step inspection when $\Delta\tau$ is small enough. Although the retailer cannot benefit from changing between these two inspection policies since its profits are zero, the single-step inspection policy is dominant to the two-step one from the social welfare perspective. Theoretically, the retailer should choose the single-step inspection policy rather than the two-step one, even though the latter seems more effective. Further, we compare the distributor's expected profits under the single-step inspection policy with those in the basic model. When the distributor uses a cold chain, the difference between them is $\frac{(1-\rho_1)\rho_2(1-\lambda)}{1-(1-\rho_1)(1-\rho_2)\lambda} \alpha(\theta_I - \theta_V)L_R - \tau$. The positive part comes from the

Table 5
Impacts of the retailer's inspection policies on the distributor's decision.

Model	Inspection policy	Inspection costs	Inspection impact
Basic Model	No inspection	/	/
Extended Models	Single-step inspection	τ	It influences the distributor towards the cold chain option.
	Two-step inspection	$\tau + \Delta\tau$	Its impact depends on the value of $\Delta\tau$; and even if it has a positive impact on the distributor's decision, it is less effective than the single-step one.

Table 6
Distributor's decisions under the retailer's inspection policies.

When $\Delta\tau > \frac{(1-\rho_1)\rho_2}{(1-(1-\rho_1)(1-\rho_2)\lambda)(1-\rho_1\beta)}\alpha(\theta_I - \theta_V)L_R$				
Distributor's Decision	Costs of a cold chain			
	$k \leq k_2$	$k_2 < k \leq k(\lambda)$	$k(\lambda) < k \leq k_1$	$k_1 < k$
No inspection	C	C	NC	NC
Single-step Inspection	C	C	C	NC
Two-step Inspection	C	NC	NC	NC
When $\Delta\tau \leq \frac{(1-\rho_1)\rho_2}{(1-(1-\rho_1)(1-\rho_2)\lambda)(1-\rho_1\beta)}\alpha(\theta_I - \theta_V)L_R$				
Distributor's Decision	Costs of a cold chain			
	$k \leq k(\lambda)$	$k(\lambda) < k \leq k_2$	$k_2 < k \leq k_1$	$k_1 < k$
No inspection	C	NC	NC	NC
Single-step Inspection	C	C	C	NC
Two-step Inspection	C	C	NC	NC

increase of the possibility of the distributor using a cold chain ($k(\lambda) < k_1$), which drops the expected adverse event loss of the retailer and then raises the vaccine price. The negative part results from the retailer shifting the inspection costs to the distributor via the vaccine price. The difference may take a positive value. In this case, we regard it as a win-win situation that the distributor uses a cold chain and meanwhile the retailer performs the single-step inspection. For the distributor, its profits are larger than those in the basic model; and for the retailer, the threshold value of the distributor's decision, i.e., the possibility of the distributor using a cold chain, is bigger than that in the basic model. When the distributor uses a non-cold chain, the difference is $-\frac{(1-\rho_1)\rho_2\lambda}{1-(1-\rho_1)(1-\rho_2)\lambda}\alpha(\theta_I - \theta_V)L_R - \tau$, which is negative. Hence, there exists no win-win situations in this case.

In addition, the discussion above proves the point that the transparent information is not always the best. Referring to the transportation situations, the asymmetric information results in the opportunities for the distributor to take advantage of it. However, the knowledge about the cold chain costs will eliminate the retailer's motive to perform an inspection, which makes its announcement incredible and useless.

4.4. Extension

We have shown that the key point of the retailer's inspection policy is to improve the retailer's ability to identify the situation where only the recording equipment malfunctions from the other situations. Next, we define $e \in [0, 1]$ as the retailer's ability (probability) to distinguish the different transportation situations for vaccines, i.e., to identify whether the distributor uses a cold chain and

which component of the cold chain malfunctions during delivery. Let $\tau(e)$ represent the retailer's inspection costs. Similarly, we have the following result (Proof: see the Appendix).

Proposition 6. Under the retailer's inspection ability e , the distributor chooses its transportation mode according to the costs of a cold chain as follows.

- (1) If $k \leq k^e(\lambda)$, the distributor uses a cold chain to transport vaccines; and
- (2) If $k > k^e(\lambda)$, the distributor uses a non-cold chain to transport vaccines,

where $k^e(\lambda) = (1 - \rho_1)\alpha(\theta_I - \theta_V)\{[1 - \frac{\rho_2(1-e)}{1-(1-\rho_1)(1-\rho_2)\lambda}]L_R + L_D\} + (1 - \rho_1)(1 - \rho_2)\tau(e)$.

We now discuss the impact of the retailer's inspection on the distributor's decision. If $\tau(e)$ is a non-decreasing function, as e increases, the motive for the distributor to choose the cold chain option increases, and the motive to choose the non-cold chain option decreases. Ideally, $e = 1$ is the best choice for the retailer. If $\tau(e)$ is a non-increasing function, there exists the somewhat counterintuitive possibility that a more effective while cheaper inspection policy decreases the distributor's motive to use a cold chain for vaccines with a small adverse event probability or loss. This finding is reminiscent of Activity Based Classification, the retailer should classify different kinds of vaccines and perform different inspection policies to them in order to balance the inspection costs and effectiveness. When the retailer has no ability to distinguish the different transportation situations, i.e., $e = 0$, Proposition 6 becomes Proposition 1. That is, Proposition 1 is the special case of Proposition 6.

Then, we further discuss the extreme threshold values with the retailer's inspection.

Proposition 7. No matter what value λ takes, we have the following results (shown in Figs. 11 and 12).

- (1) If $k \leq k_3$, the distributor uses a cold chain to transport vaccines; and
- (2) If $k > k_4$, the distributor uses a non-cold chain to transport vaccines,

Where $k_3 = (1 - \rho_1)\alpha(\theta_I - \theta_V)[\frac{\rho_1(1-\rho_2)+e\rho_2}{\rho_1+\rho_2-\rho_1\rho_2}L_R + L_D] + (1 - \rho_1)(1 - \rho_2)\tau(e)$ and $k_4 = (1 - \rho_1)\alpha(\theta_I - \theta_V)[(1 - \rho_2 + e\rho_2)L_R + L_D] + (1 - \rho_1)(1 - \rho_2)\tau(e)$.

Comparing k_3 with k' and k_4 with k'' shows that $k_3 \geq k'$ and $k_4 \geq k''$. The result indicates that the retailer's inspection makes the extreme threshold values of the distributor's decision move to the right. We propose two reasons for this impact. First, it is understandable that the retailer will transfer its inspection costs to

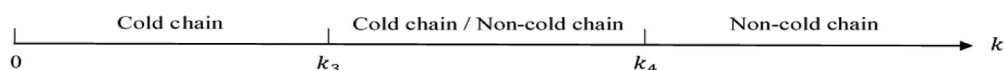


Fig. 11. Extreme threshold values under the retailer's inspection ability.

Notes: Example for $\tau(e) = e^2/2$, $\rho_1 = \rho_2 = 0.2$, $L = 2$, $\alpha = 0.8$, $\theta_I = 0.5$, $\theta_V = 0.1$.

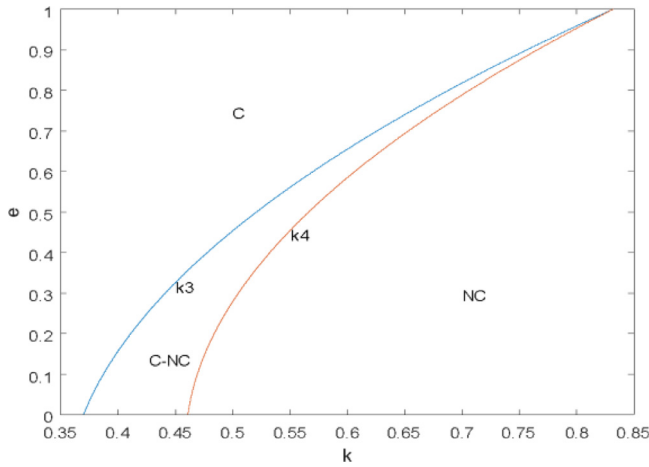


Fig. 12. Distributor's decisions in the $k-e$ space under the retailer's inspection ability.

the vaccine price, which indirectly affects the extreme threshold values. Second, comparing the first part of k' and k_3 , the difference between them is $\frac{e(1-\rho_1)\rho_2}{\rho_1+\rho_2-\rho_1\rho_2}L_R$. When the retailer totally trusts the distributor, the situation where only the recording equipment malfunctions happens according to the probability $(1-\rho_1)\rho_2$, and the whole probability of the transportation result being no data is $\rho_1+\rho_2-\rho_1\rho_2$. The retailer forms the belief about valid vaccines using Bayes' rule. When the retailer performs an inspection, it will identify the recording equipment malfunction situation with the probability e . Hence, the distributor relaxes its tolerance range of the cold chain costs by $\frac{e(1-\rho_1)\rho_2}{\rho_1+\rho_2-\rho_1\rho_2}L_R$ due to the retailer's inspection. The retailer's inspection effects the extreme threshold value of the distributor's decision and then the retailer's expected adverse event loss. We can also regard the difference as a loss shift from the retailer to the distributor. According to Proposition 3, such a shift will increase the threshold values. Similarly, we can analyze k'' and k_4 .

5. Conclusion

In this study, we focus on the distributor's cold chain transportation decision problem of temperature-sensitive vaccines in developing countries. We examine the distributor's motives to transport vaccines using a cold chain or non-cold chain and analyze how the retailer's inspection affects the distributor's decision-making in its transportation mode for vaccines. By establishing and using the basic model, we determine under what conditions the distributor will use a cold chain or non-cold chain to transport vaccines. Further, two commonly used inspection policies of the retailer are introduced into the basic model. First, we find that the retailer's single-step inspection influences the distributor to choose the cold chain option. Second, the comparison between these two inspection policies shows that the two-step inspection is less effective than the single-step one in this influence. These results suggest that the retailer should improve their inspection policy in order to identify the distributor's decision and the malfunctions situations. We also show that the asymmetric information of the cold chain costs between the two parties has a positive effect on ensuring the retailer to perform an inspection.

In the future, some follow-up studies could be conducted to further discuss the impact of the retailer's inspection on the distributor's decision considering a repeated game between the two parties and the retailer's random inspection policy. In this paper, we assume that the retailer will accept the vaccines that are transported using a non-cold chain when its income from the vaccines

will cover its expected loss in a vaccine-related adverse event and the vaccine price. Such an assumption may lead to collusion between the distributor and the retailer. It is not enough to only depend on the retailer to prevent invalid vaccines from entering the market. It will make sense to explore how policy makers regulate and improve the distributor's and the retailer's behavior using measures, such as regulation frequency, penalty or reward.

Acknowledgments

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Appendix

Proof of Proposition 1

In this proposition, we discuss under what condition the distributor will choose the cold chain or non-cold chain option. Here, the retailer's trust level in the distributor to use a cold chain for vaccines is within the interval $[0, 1]$, i.e., $\lambda \in [0, 1]$.

When the distributor uses a cold chain to transport vaccines ($t=C$), the transportation result will be data ($r=D$) with the probability $(1-\rho_1)(1-\rho_2)$ and be no data ($r=ND$) with the complementary probability. If $r=D$, the retailer's belief about valid vaccines is α ($\mu=\alpha$); and if $r=ND$, the retailer forms the belief using Bayes' rule.

$$\mu_D = \alpha$$

$$\mu_{ND} = \frac{(1-\rho_1)\rho_2\lambda}{[1-(1-\rho_1)(1-\rho_2)]\lambda + 1*(1-\lambda)}\alpha$$

According to the basic expression of the vaccine price $p = v - [\mu\theta_V + (1-\mu)\theta_I]L_R$, its specific expression can be written as

$$p_D = v - [\mu_D\theta_V + (1-\mu_D)\theta_I]L_R$$

$$p_{ND} = v - [\mu_{ND}\theta_V + (1-\mu_{ND})\theta_I]L_R$$

The price difference is

$$p_D - p_{ND} = (\mu_D - \mu_{ND})(\theta_I - \theta_V)L_R$$

The distributor's expected profits can be written as

$$\pi_C^D = (1-\rho_1)(1-\rho_2)p_D + [1-(1-\rho_1)(1-\rho_2)]p_{ND}$$

$$- k - \{(1-\rho_1)\alpha\theta_V + [1-(1-\rho_1)\alpha]\theta_I\}L_D$$

When the distributor uses a non-cold chain for vaccines ($t=NC$), it definitely cannot provide the necessary temperature data to the retailer ($r=ND$). In this case, the distributor's expected profits are

$$\pi_{NC}^D = p_{ND} - \theta_I L_D$$

Overall, we get the following:

$$\pi_C - \pi_{NC} = (1-\rho_1)(1-\rho_2)(p_D - p_{ND})$$

$$+ (1-\rho_1)\alpha(\theta_I - \theta_V)L_D - k$$

Apparently, the distributor's motive to transport vaccines using a cold chain decreases with the decreasing price difference. Note that $\theta_V < \theta_I$. It is easy to prove that the price difference strictly decreases with the increasing retailer's trust level.

Based on the discussion above, the distributor will choose the cold chain option if $\pi_C^D \geq \pi_{NC}^D$, i.e., the costs of a cold chain $k \leq k(\lambda) = (1-\rho_1)\alpha(\theta_I - \theta_V)\{\frac{(1-\rho_2)[1-(1-\rho_1)\lambda]}{1-(1-\rho_1)(1-\rho_2)\lambda}L_R + L_D\}$.

Proof of Proposition 2

(1) Cold chain decision

We now discuss under what condition the distributor will definitely choose the cold chain option. Assume that the retailer relies

on the distributor to use a cold chain for vaccines. In other words, the trust level of the retailer in the distributor is one, i.e., $\lambda = 1$.

When the distributor uses a cold chain to transport vaccines ($t = C$), the transportation result will be data or no data. When $r = D$, the retailer's belief about valid vaccines μ will be its prior belief α ; when $r = ND$, the retailer will revise its belief using Bayes' rule.

$$\mu_D = \alpha$$

$$\mu_{ND} = \frac{(1-\rho_1)\rho_2\alpha}{\rho_1+\rho_2-\rho_1\rho_2}$$

Therefore, the vaccine price is

$$p_D = v - [\mu_D\theta_V + (1 - \mu_D)\theta_I]L_R$$

$$p_{ND} = v - [\mu_{ND}\theta_V + (1 - \mu_{ND})\theta_I]L_R$$

Therefore, we can get the distributor's expected profits, which is written as

$$\pi_C^D = (1 - \rho_1)(1 - \rho_2)p_D + [1 - (1 - \rho_1)(1 - \rho_2)]p_{ND}$$

$$- k - \{(1 - \rho_1)\alpha\theta_V + [1 - (1 - \rho_1)\alpha]\theta_I\}L_D$$

When the distributor does not use a cold chain to transport vaccines, the distributor will not trade at the actual non-cold chain vaccine price, since it is lower than p_{ND} , while the retailer cannot distinguish the non-cold chain situation from the malfunctions. We then get the distributor's expected profits

$$\pi_{NC}^D = p_{ND} - \theta_I L_D$$

Based on the discussion, the distributor will choose the cold chain option if $\pi_C^D \geq \pi_{NC}^D$, i.e., the cost of a cold chain $k \leq k' = (1 - \rho_1)\alpha(\theta_I - \theta_V)[\frac{\rho_1(1-\rho_2)}{\rho_1+\rho_2-\rho_1\rho_2}L_R + L_D]$.

(2) Non-cold chain decision

We next discuss under what condition the distributor will definitely choose the non-cold chain option. Similarly, we set the retailer's trust level in the distributor to use a cold chain for vaccines as zero, i.e., $\lambda = 0$.

When the distributor uses a cold chain to transport vaccines, if $r = D$, the retailer's belief about valid vaccines is α , i.e., $\mu = \alpha$; and if $r = ND$, it turns to be zero, $\mu = 0$.

$$\mu_D = \alpha$$

$$\mu_{ND} = 0$$

The vaccine price is

$$p_D = v - [\mu_D\theta_V + (1 - \mu_D)\theta_I]L_R$$

$$p_{ND} = v - [\mu_{ND}\theta_V + (1 - \mu_{ND})\theta_I]L_R$$

Next, we can get the distributor's expected profits, which is written as

$$\pi_C^D = (1 - \rho_1)(1 - \rho_2)p_D + [1 - (1 - \rho_1)(1 - \rho_2)]p_{ND}$$

$$- k - \{(1 - \rho_1)\alpha\theta_V + [1 - (1 - \rho_1)\alpha]\theta_I\}L_D$$

When the distributor does not use a cold chain to transport vaccines, similarly, we can get the expected profits of the distributor, which is written as

$$\pi_{NC}^D = p_{ND} - \theta_I L_D$$

Therefore, the distributor will choose the non-cold chain option if $\pi_C^D \leq \pi_{NC}^D$, i.e., the costs of a cold chain $k \geq k'' = (1 - \rho_1)\alpha(\theta_I - \theta_V)[(1 - \rho_2)L_R + L_D]$.

Proof of Proposition 3

We let $\eta = L_D/L$ denote the ratio of the distributor's loss to the full loss of a vaccine-related adverse event. Let $\Delta = \theta_I - \theta_V$ denote the difference between the accident probabilities of the two types

of vaccines. Then, $k(\lambda)$, k' and k'' can be written as

$$k(\lambda) = (1 - \rho_1) \left\{ \frac{(1 - \rho_2)[1 - (1 - \rho_1)\lambda]}{1 - (1 - \rho_1)(1 - \rho_2)\lambda} \frac{1 - \eta}{\eta} + 1 \right\} L_D \alpha \Delta$$

$$k' = (1 - \rho_1) \left[\frac{(1 - \rho_2)\rho_1}{\rho_1 + \rho_2 - \rho_1\rho_2} \frac{1 - \eta}{\eta} + 1 \right] L_D \alpha \Delta$$

$$k'' = (1 - \rho_1) \left[(1 - \rho_2) \frac{1 - \eta}{\eta} + 1 \right] L_D \alpha \Delta$$

It's also easy to prove that they increases strictly in η and Δ .

Proof of Proposition 4

Under the retailer's single-step inspection policy, when the distributor uses a cold chain to transport vaccines, there are two possible inspection results, pass (P) or fail (F), which are represented by $i \in \{P, F\}$. Once a malfunction occurs in the controlling equipment, the distributor will fail the inspection according to the probability β , and pass with the complementary value $1 - \beta$. Then, we can get

$$\mu_P = \frac{(1-\rho_1)\alpha}{(1-\rho_1)+\rho_1(1-\beta)} \mu_F = 0$$

$$\mu_F = 0$$

According to the extended expression of the vaccine price $p = v - [\mu\theta_V + (1 - \mu)\theta_I]L_R - \tau$, its specific expression can be written as

$$p_P = v - [\mu_P\theta_V + (1 - \mu_P)\theta_I]L_R - \tau$$

$$p_F = v - [\mu_F\theta_V + (1 - \mu_F)\theta_I]L_R - \tau$$

Then we can write the expected profits of the distributor as

$$\pi_C^D = [(1 - \rho_1) + \rho_1(1 - \beta)]p_P + \rho_1\beta p_F$$

$$- k - \{(1 - \rho_1)\alpha\theta_V + [1 - (1 - \rho_1)\alpha]\theta_I\}L_D$$

When the distributor does not to use a cold chain, the inspection fails. The expected profits of the distributor are

$$\pi_{NC}^D = p_F - \theta_I L_D$$

Therefore, the distributor will choose the cold chain option if $\pi_C^D \geq \pi_{NC}^D$, i.e., the costs of a cold chain $k \leq k_1 = (1 - \rho_1)\alpha(\theta_I - \theta_V)L$.

Proof of Proposition 5

Similarly, under the retailer's two-step inspection policy, when the distributor uses a cold chain to transport vaccines, i.e., $t = C$, we can get

$$\mu_{(P,P)} = \alpha$$

$$\mu_{(P,F)} = \frac{(1 - \rho_1)\rho_2\alpha}{(1 - \rho_1)\rho_2 + \rho_1(1 - \beta)}$$

$$\mu_F = 0$$

Then the vaccine price can be written as

$$p_{(P,P)} = v - [\mu_{(P,P)}\theta_V + (1 - \mu_{(P,P)})\theta_I]L_R - (\tau + \Delta\tau)$$

$$p_{(P,F)} = v - [\mu_{(P,F)}\theta_V + (1 - \mu_{(P,F)})\theta_I]L_R - (\tau + \Delta\tau)$$

$$p_F = v - [\mu_F\theta_V + (1 - \mu_F)\theta_I]L_R - \tau$$

Next, we can write the expected profits of the distributor as

$$\pi_C^D = (1 - \rho_1)(1 - \rho_2)p_{(P,P)} + [(1 - \rho_1)\rho_2 + \rho_1(1 - \beta)]p_{(P,F)}$$

$$+ \rho_1\beta p_F - k - \{(1 - \rho_1)\alpha\theta_V + [1 - (1 - \rho_1)\alpha]\theta_I\}L_D$$

When the distributor does not to use a cold chain, the inspection result is F . The expected profits of the distributor are

$$\pi_{NC}^D = p_F - \theta_I L_D$$

Therefore, the distributor will choose the cold chain option if $\pi_C^D \geq \pi_{NC}^D$, i.e., the costs of a cold chain $k \leq k_2 = (1 - \rho_1)\alpha(\theta_l - \theta_v)L - (1 - \rho_1)\beta)\Delta\tau$.

Proof of Proposition 6

Under the retailer's inspection ability, we discuss the distributor's cold chain transportation decision. Here, the retailer's trust level in the distributor to use a cold chain for vaccines is within the interval $[0, 1]$, i.e., $\lambda \in [0, 1]$.

When the transportation result is data ($r = D$), the retailer's belief about valid vaccines will be α ; when the transportation result is no data ($r = ND$) and the distributor cannot identify the transportation situation, the retailer forms the belief using Bayes' rule; and when the transportation result is no data and the distributor can identify the transportation situation, the belief will be α in the recording equipment malfunction situation or zero in the other situations.

$$\begin{aligned}\mu_D &= \alpha \\ \mu_{ND} &= \frac{(1-\rho_1)\rho_2\lambda\alpha}{[1-(1-\rho_1)(1-\rho_2)]\lambda+1*(1-\lambda)} \\ \mu_{NC} &= 0\end{aligned}$$

Therefore, the vaccine price is

$$\begin{aligned}p_D &= v - [\mu_D\theta_v + (1 - \mu_D)\theta_l]L_R \\ p_{ND} &= v - [\mu_{ND}\theta_v + (1 - \mu_{ND})\theta_l]L_R - \tau(e) \\ p_{NC} &= v - [\mu_{NC}\theta_v + (1 - \mu_{NC})\theta_l]L_R - \tau(e)\end{aligned}$$

When the distributor uses a cold chain to transport vaccines, the distributor's expected profits can be written as

$$\begin{aligned}\pi_C^D &= (1 - \rho_1)(1 - \rho_2)p_D + (1 - \rho_1)\rho_2[e(p_D - \tau(e)) \\ &\quad + (1 - e)p_{ND}] + \rho_1[ep_{NC} + (1 - e)p_{ND}] \\ &\quad - k - \{(1 - \rho_1)\alpha\theta_v + [1 - (1 - \rho_1)\alpha]\theta_l\}L_D\end{aligned}$$

When the distributor uses a non-cold chain for vaccines, the distributor's expected profits are

$$\pi_{NC}^D = ep_{NC} + (1 - e)p_{ND} - \theta_l L_D$$

The distributor will choose the cold chain option if $\pi_C^D \geq \pi_{NC}^D$, i.e., the costs of a cold chain $k \leq k(\lambda) = (1 - \rho_1)\alpha(\theta_l - \theta_v)[\{1 - \frac{\rho_2(1-e)}{1-(1-\rho_1)(1-\rho_2)\lambda}\}L_R + L_D] + (1 - \rho_1)(1 - \rho_2)\tau(e)$.

Proof of Proposition 7

Cold chain decision

Here, we assume that the retailer entirely trusts the distributor to use a cold chain for vaccines, i.e., $\lambda = 1$.

When the distributor provides the necessary temperature data to the retailer on delivery, we can get

$$\begin{aligned}\mu_D &= \alpha \\ p_D &= v - [\mu_D\theta_v + (1 - \mu_D)\theta_l]L_R\end{aligned}$$

When the distributor does not provide the data and the retailer cannot identify the malfunctions, we can get

$$\begin{aligned}\mu_{ND} &= \frac{(1-\rho_1)\rho_2\alpha}{\rho_1+\rho_2-\rho_1\rho_2} \\ p_{ND} &= v - [\mu_{ND}\theta_v + (1 - \mu_{ND})\theta_l]L_R - \tau(e)\end{aligned}$$

When the distributor does not provide the data and the retailer can identify the malfunctions, we have the following analysis. When the retailer finds that only the recording equipment breaks down, the belief about valid vaccines will be μ_D , and the vaccine price will be p_D ; when the retailer finds that the controlling equipment malfunctions during transportation or that the distributor uses a non-cold chain for vaccines, we can get

$$\begin{aligned}\mu_{NC} &= 0 \\ p_{NC} &= v - [\mu_{NC}\theta_v + (1 - \mu_{NC})\theta_l]L_R - \tau(e)\end{aligned}$$

Therefore, we can get the distributor's expected profits, which is written as

$$\begin{aligned}\pi_C^D &= (1 - \rho_1)(1 - \rho_2)p_D + (1 - \rho_1)\rho_2[e(p_D - \tau(e)) \\ &\quad + (1 - e)p_{ND}] + \rho_1[ep_{NC} + (1 - e)p_{ND}] \\ &\quad - k - \{(1 - \rho_1)\alpha\theta_v + [1 - (1 - \rho_1)\alpha]\theta_l\}L_D\end{aligned}$$

$$\pi_{NC}^D = ep_{NC} + (1 - e)p_{ND} - \theta_l L_D$$

The distributor will choose the cold chain option if $\pi_C^D \geq \pi_{NC}^D$, i.e., the costs of a cold chain $k \leq k_3 = (1 - \rho_1)\alpha(\theta_l - \theta_v)[\frac{\rho_1(1-\rho_2)+e\rho_2}{\rho_1+\rho_2-\rho_1\rho_2}L_R + L_D] + (1 - \rho_1)(1 - \rho_2)\tau(e)$.

(2) Non-cold chain decision

We next discuss under what condition the distributor will choose the non-cold chain option. Here, the retailer does not trust the distributor to use a cold chain for vaccines at all, i.e., $\lambda = 0$. Then, we can get

$$\begin{aligned}\pi_C^D &= (1 - \rho_1)(1 - \rho_2)p_D + (1 - \rho_1)\rho_2[e(p_D - \tau(e)) \\ &\quad + (1 - e)p_{NC}] + \rho_1[ep_{NC} + (1 - e)p_{NC}] \\ &\quad - k - \{(1 - \rho_1)\alpha\theta_v + [1 - (1 - \rho_1)\alpha]\theta_l\}L_D\end{aligned}$$

$$\pi_{NC}^D = ep_{NC} + (1 - e)p_{NC} - \theta_l L_D$$

The distributor will choose the non-cold chain option if $\pi_C^D \leq \pi_{NC}^D$, i.e., the costs of a cold chain $k > k_4 = (1 - \rho_1)\alpha(\theta_l - \theta_v)[(1 - \rho_2 + \rho_2e)L_R + L_D] + (1 - \rho_1)(1 - \rho_2)\tau(e)$.

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